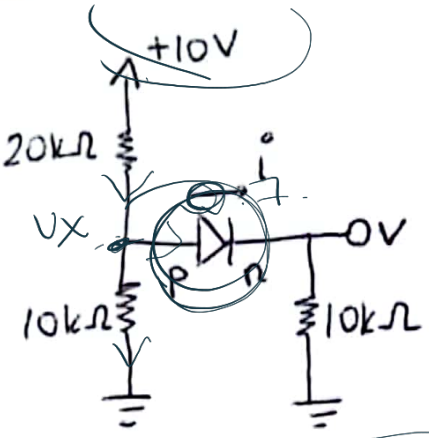
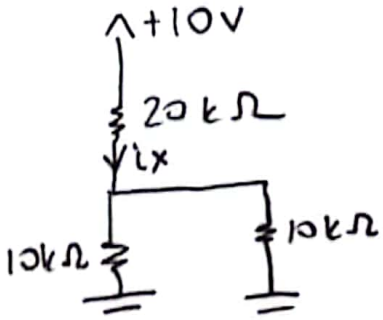
Quiz 1 Sample

Q1

$V_{D0} = 0.7V$ Find the i value.



Diode ⇒ on



$\frac{10 - 2.28}{10}$

$\frac{10 - V_x}{20k} = \frac{V_x}{10k} + \frac{V_x - 0.7}{10k}$

(2) (2)

$i_x = \frac{10V}{25k\Omega} = 0.4mA$

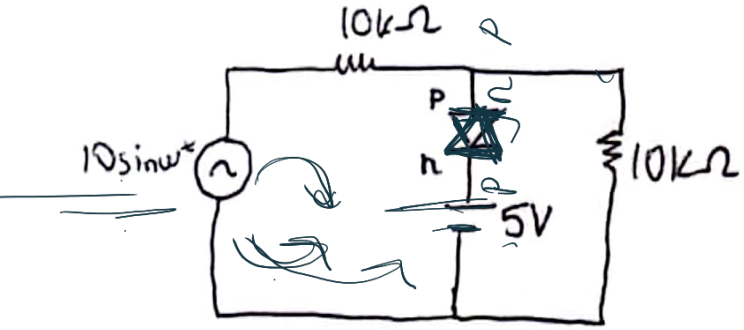
$Req = 20 + 5 = 25k\Omega$

$0.4mA \cdot \frac{10}{20} = 0.2mA$

$10 - V_x = 2V_x \neq 2V_x - 1.4$

$\frac{11.4}{5} = 2.28\ V_x$

Q2 $11.4 = 5V_x$



draw the wave form V_D

$V_{D0} = 0.7V$

$V_p = 10\sin\omega t \quad V_n = 5V$

$V_D = 10\sin\omega t - 5V \geq V_{D0}$

$5 - 10\sin\omega t \geq 0.7$

$10\sin\omega t \geq 4.3$

0.7V

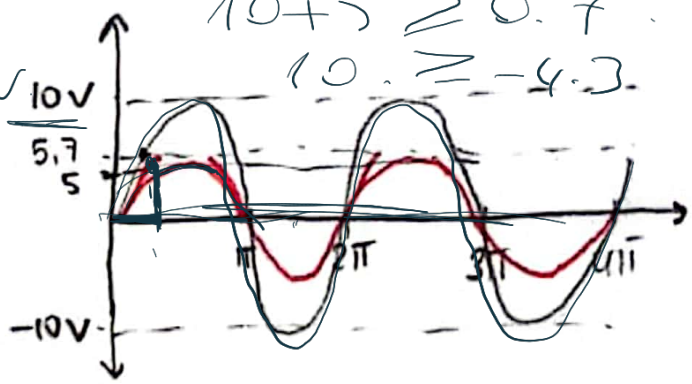
diode ⇒ on

0.7

$10 + 5 \geq 0.7$

$10 \geq -4.3$

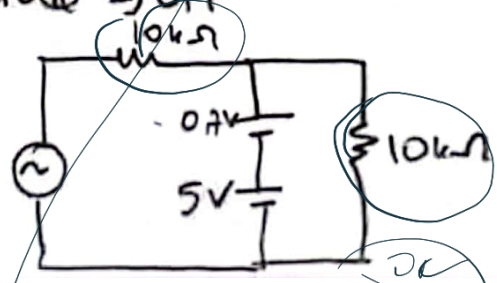
5.7V



till sins voltage reaches 5.7

diode ⇒ off

But voltage will never reach 5.7 volt so diode will never be on.

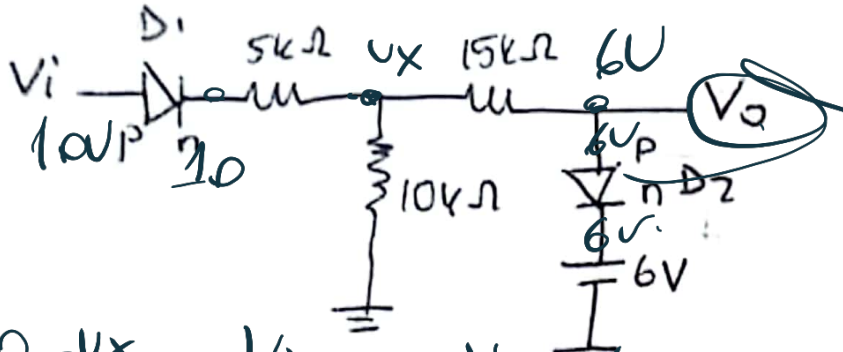


$5 = 10\sin\omega t \Rightarrow V_D \geq 0.7$

$10\sin\omega t + 1.4 \geq 0.7$

CamScanner ile tarandı

Q3 — 10 sin ωt) — 4.3



Vi	Vo
-3V	0V
0V	0V
3V	2V
10V	6V

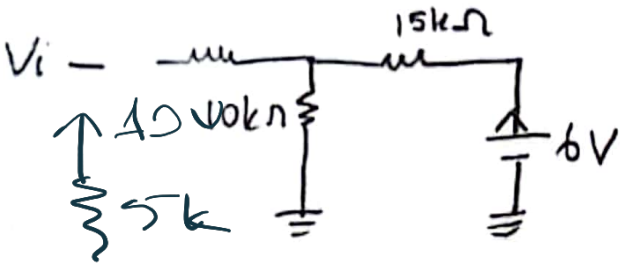
$$\frac{10 - V_x}{5k} = \frac{V_x}{10} + \frac{V_x - 6}{15}$$

(6) (3) (2)

Find the output voltages for each Vi. Use ideal model.

Vi = -3V

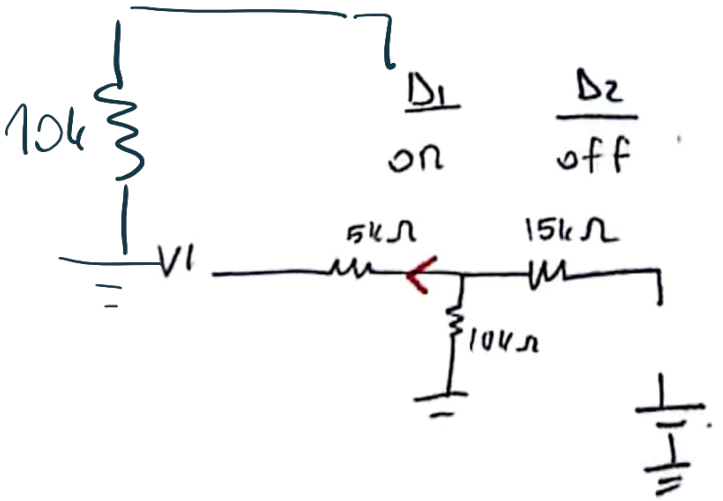
D1	D2
off	on



D1	D2
ON	ON

$$\frac{6V}{25k\Omega} = 0.24mA$$

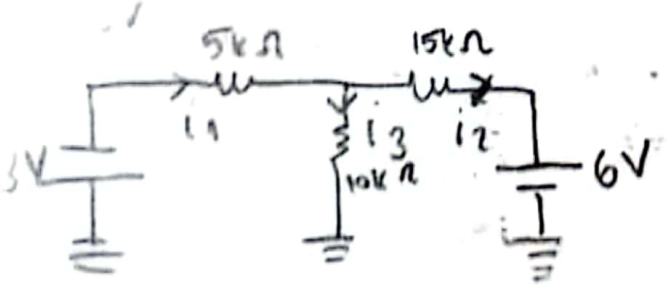
current wrong direct. X



X current wrong direc.

D1	D2
on	on

X



$i_1 = i_2 + i_3$

$-10ki_3 + 15ki_2 + 6 = 0$ (i7 = -)

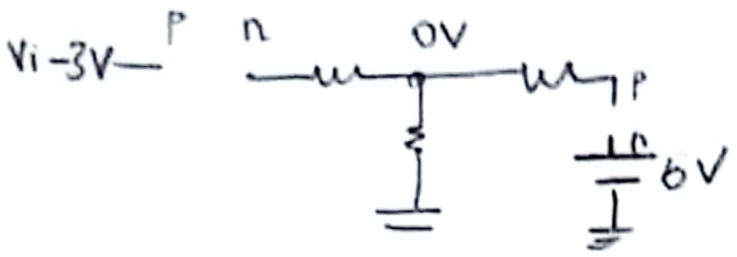
$10i_3 - 15i_2 = 6$ $55i_3 = -3$ $i_3 = -$

$3 + 5i_1 + 10i_3 = 0$

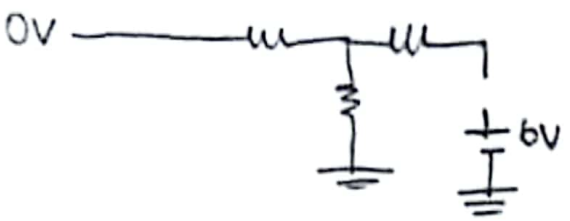
$3 + 15i_2 + 15i_2 = 0$ $3 + 15i_2 + 10(5i_3) = 0$

CamScanner ile tarandı

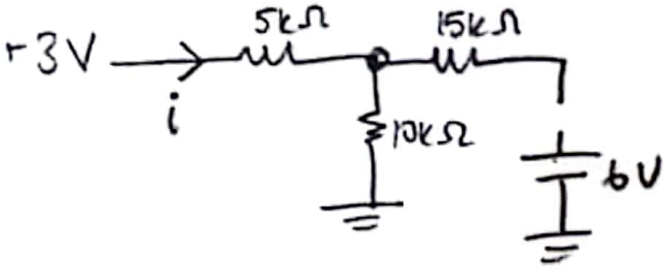
D_1 off D_2 off ✓



$V_i = 0V$ D_1 on D_2 off ✓



$V_i = 3V$ D_1 on D_2 off ✓

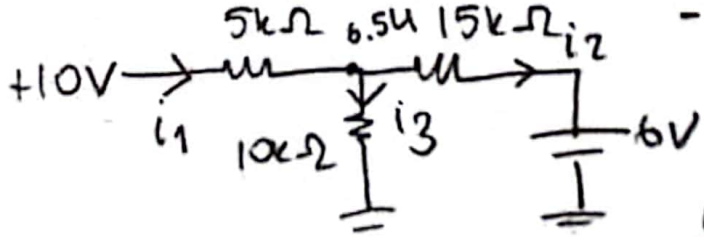


$$i = \frac{3V}{15k\Omega} = 0.2mA$$

$$V_o = 3 - 0.2mA \times 5k$$

$$V_o = 2V$$

$V_i = 10V$ D_1 on D_2 on ✓



$i_1 = i_2 + i_3$

$-10i_3 + 15i_2 + 6 = 0$

$-10 + 5i_1 + 10i_3 = 0$

$5i_2 + 15i_3 = 20$

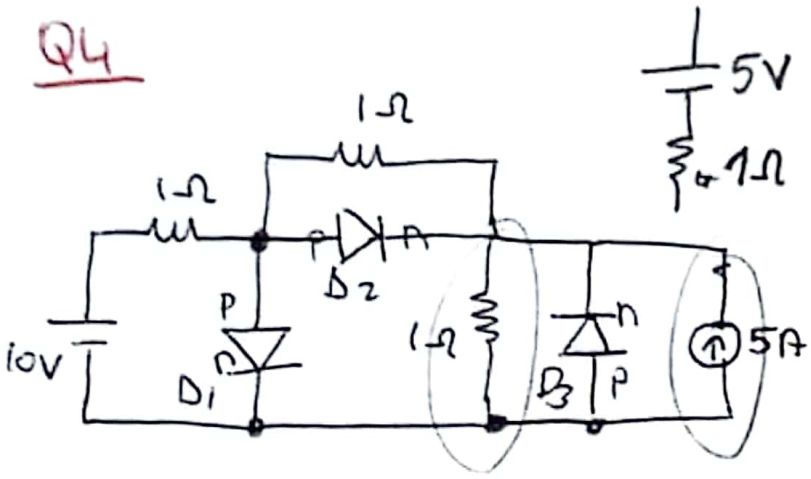
$15i_2 - 10i_3 = -6$

$55i_2 = 2$ $i_2 = 0.036$

$i_1 = 0.69$ $i_3 = 0.65$

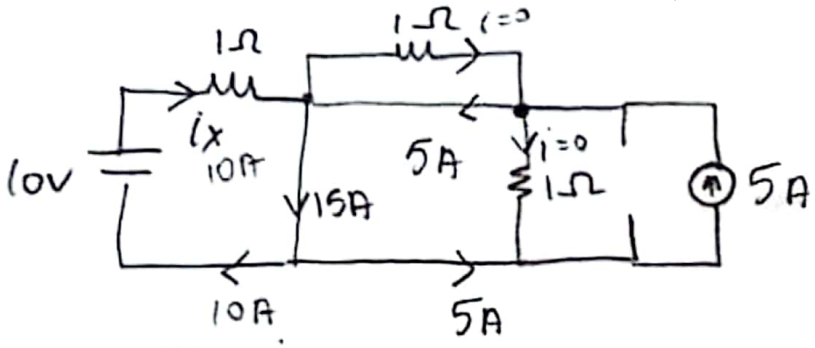
CamScanner ile tarandı

Q4



Find the currents in each diode. (ideal diode model)

D1	D2	D3
on	on	off

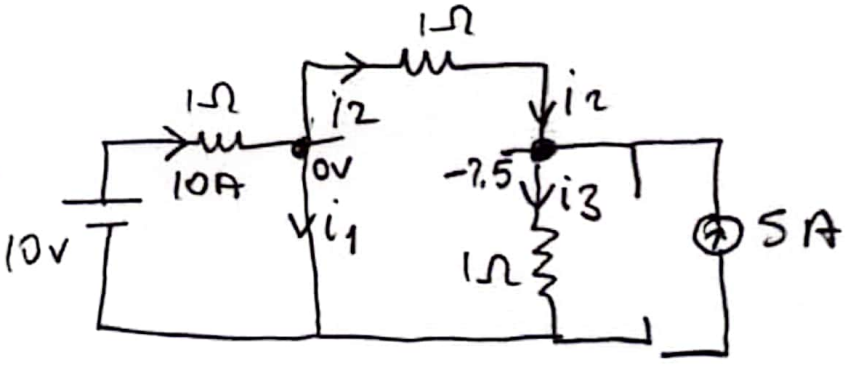


$-10 + i_x = 0$

D2 current wrong ✗

D1	D2	D3
on	off	off

✗



$V_{D3} = 0 - -2.5 = 2.5$

$V_{D2} = 0 - -2.5 = 2.5$

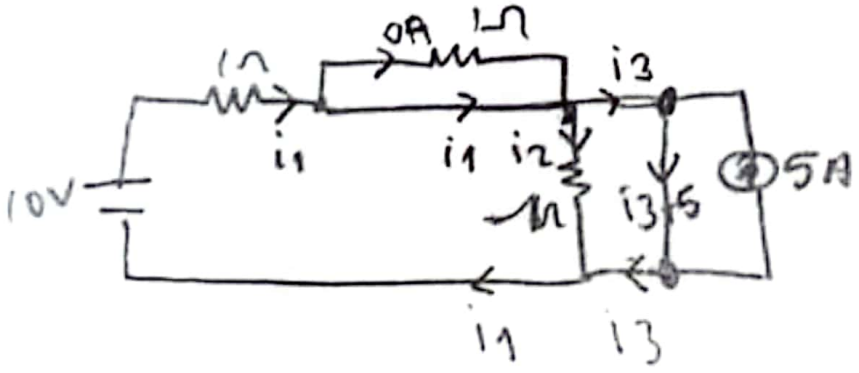
$i_1 + i_2 = 10A \quad i_2 + 5A = i_3$

$i_2 + i_3 = 0 \quad i_2 = -2.5A$

$i_2 - i_3 = -5 \quad i_3 = 2.5A$

$i_1 = 12.5A$

D1	D2	D3
off	on	on



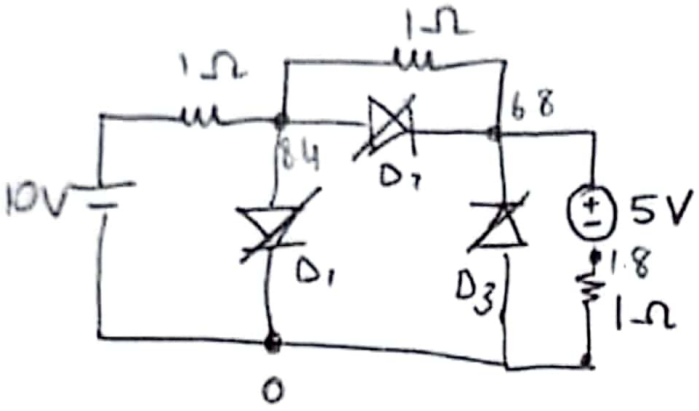
$i_1 = i_2 + i_3$

$-10 + i_1 + i_2 = 0 \quad i_2 = 0 \quad i_1 = 10$

$i_1 + i_2 = 10 \quad 2i_1 = 10 + i_3$

$i_1 - i_2 = i_3 \quad i_3 = 10$

CamScanner ile tarandı



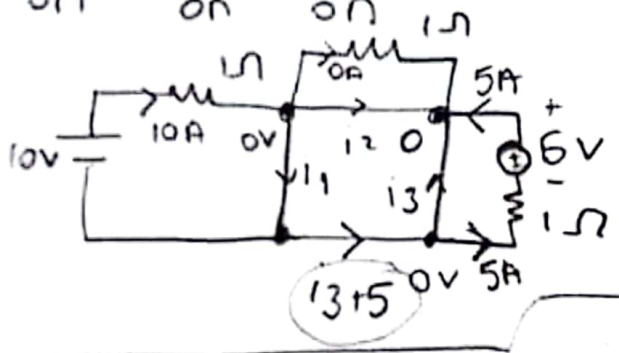
D1	D2	D3
off	off	off

$i = \frac{10-5}{3} = 1.6$

$V_{D_1} = 8.4$ ✗

D1	D2	D3
on	on	on

✗



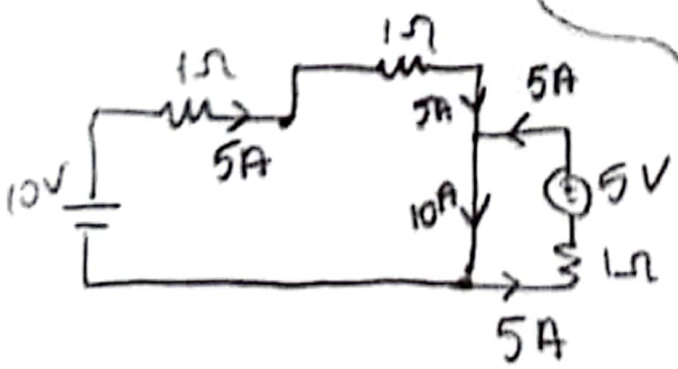
$i_1 + i_2 = 10A$

$i_2 + i_3 + 5 = 0$

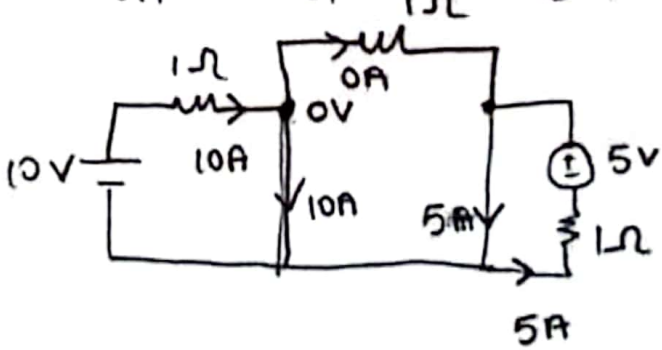
i_2 or i_3 "−"

D1	D2	D3
✓on	on	on
✓on	on	of
✓on	off	on
✓on	off	off
✓off	on	on
off	on	off
off	off	on
✓off	off	off

D1	D2	D3
off	off	on



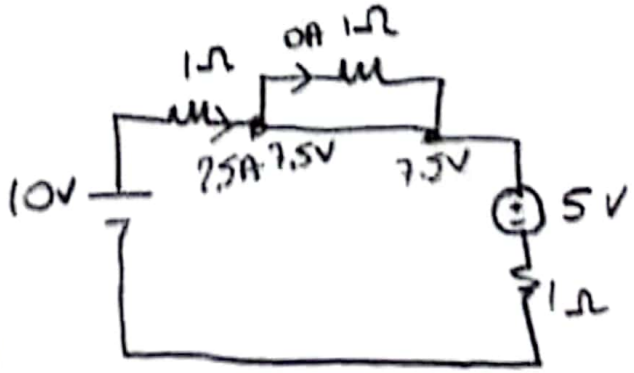
D1	D2	D3
on	off	on



5A wrong direction

✗

D1	D2	D3
off	on	off



$\frac{10-5}{2} = 2.5$

CamScanner ile tarandı